```
natavan@DESKTOP-GSHQS04:~$ mkdir cafe-intermediate && cd cafe-intermediate
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ touch index.html styles.css
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ ls -la
total 0
drwxr-xr-x 1 natavan natavan 512 Mar 31 15:43 .
drwxr-x--- 1 natavan natavan 512 Mar 31 15:42 ..
-rw-r--r-- 1 natavan natavan   0 Mar 31 15:43 index.html
-rw-r--r-- 1 natavan natavan   0 Mar 31 15:43 styles.css
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/natavan/cafe-intermediate/.git/
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ ls -la
total 0
drwxr-xr-x 1 natavan natavan 512 Mar 31 15:44 .
drwxr-x--- 1 natavan natavan 512 Mar 31 15:42 ..
drwxr-xr-x 1 natavan natavan 512 Mar 31 15:44 .git
-rw-r--r-- 1 natavan natavan   0 Mar 31 15:43 index.html
-rw-r--r-- 1 natavan natavan   0 Mar 31 15:43 styles.css
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git add .
```

```
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git commit -m "chore: project scaffold (index & css)"
[master (root-commit) f679976] chore: project scaffold (index & css)
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 index.html
 create mode 100644 styles.css
```

```
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git remote add origin https://github.com/natavann/Cafe-Intermediate.git
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git branch -m main
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git push -u origin main
Username for 'https://github.com': natavann
Password for 'https://natavann@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 246 bytes | 82.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/natavann/Cafe-Intermediate.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Exercise 1 — `feature/about-page`

```
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git checkout -b feature/about-page
Switched to a new branch 'feature/about-page'
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ echo "<h1>About Café Git</h1>" > about.html
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git add about.html
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git commit -m "feat: add bare about.html"
[feature/about-page 82c3e54] feat: add bare about.html
 1 file changed, 1 insertion(+)
 create mode 100644 about.html
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git push -u origin feature/about-page
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 320 bytes | 160.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'feature/about-page' on GitHub by visiting:
remote:      https://github.com/natavann/Cafe-Intermediate/pull/new/feature/about-page
remote:
To https://github.com/natavann/Cafe-Intermediate.git
 * [new branch]      feature/about-page -> feature/about-page
branch 'feature/about-page' set up to track 'origin/feature/about-page'.
```

Exercise 2 — `feature/contact-page`

```
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git pull
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 910 bytes | 455.00 KiB/s, done.
From https://github.com/natavann/Cafe-Intermediate
   f679976..20184c1  main       -> origin/main
Updating f679976..20184c1
Fast-forward
 about.html | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 about.html
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git checkout -b feature/contact-page
Switched to a new branch 'feature/contact-page'
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ echo "<h1>About Café Git</h1>" > contact.html
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git add contact.html
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git commit -m "feat: add bare contact.html"
[feature/contact-page f32ab7b] feat: add bare contact.html
 1 file changed, 1 insertion(+)
 create mode 100644 contact.html
```

```
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git push -u origin feature/contact-page
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 237 bytes | 237.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature/contact-page' on GitHub by visiting:
remote:      https://github.com/natavann/Cafe-Intermediate/pull/new/feature/contact-page
remote:
To https://github.com/natavann/Cafe-Intermediate.git
 * [new branch]      feature/contact-page -> feature/contact-page
branch 'feature/contact-page' set up to track 'origin/feature/contact-page'.
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$
```

Exercise 3 — Merge Conflict Drill

```
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git checkout -b text-change-A
Switched to a new branch 'text-change-A'
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ echo "<p>Best coffee in town ☕</p>" >> index.html
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git add index.html
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git commit -m "feat: slogan variant A"
[text-change-A cbe16b2] feat: slogan variant A
 1 file changed, 1 insertion(+)
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git push -u origin text-change-A
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 292 bytes | 292.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'text-change-A' on GitHub by visiting:
remote:      https://github.com/natavann/Cafe-Intermediate/pull/new/text-change-A
remote:
To https://github.com/natavann/Cafe-Intermediate.git
 * [new branch]      text-change-A -> text-change-A
branch 'text-change-A' set up to track 'origin/text-change-A'.
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git checkout -b text-change-B
Switched to a new branch 'text-change-B'
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ echo "<p>Your daily dose of caffeine</p>" >> index.html
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git add index.html
```

```
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git add index.html
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git commit -m "feat: slogan variant B"
[text-change-B f014ac3] feat: slogan variant B
 1 file changed, 1 insertion(+)
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git push -u origin text-change-B
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 349 bytes | 349.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'text-change-B' on GitHub by visiting:
remote:      https://github.com/natavann/Cafe-Intermediate/pull/new/text-change-B
remote:
To https://github.com/natavann/Cafe-Intermediate.git
 * [new branch]      text-change-B -> text-change-B
branch 'text-change-B' set up to track 'origin/text-change-B'.
```

index.html 1 conflict Prev

Accept current change | Accept incoming change | Accept both changes

```
1 <<<<<<< text-change-B (Current change)
2 <p>Your daily dose of caffeine</p>
3 =======
4 <p>Best coffee in town ☕</p>
5 >>>>>>> main (Incoming change)
6
```

Exercise 4 — Stash & Dash

```
atavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git add contact.html && git commit -m "fix: typo in contact"
fatal: pathspec 'contact.html' did not match any files
atavan@DESKTOP-GSHQS04:~/cafe-intermediate$ nano styles.css
atavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git stash push -m "wip: colour tweak"
aved working directory and index state On main: wip: colour tweak
atavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git stash list
tash@{0}: On main: wip: colour tweak
atavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git checkout main
lready on 'main'
our branch is up to date with 'origin/main'.
atavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git stash push -m "wip: colour tweak"^C
atavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git checkout main
lready on 'main'
our branch is up to date with 'origin/main'.
atavan@DESKTOP-GSHQS04:~/cafe-intermediate$ echo "<!-- fixed typo -->" >> contact.html
bash: !-: event not found
atavan@DESKTOP-GSHQS04:~/cafe-intermediate$ nano contact.html
atavan@DESKTOP-GSHQS04:~/cafe-intermediate$
atavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git add contact.html && git commit -m "fix: typo in contact"
main e2760a1] fix: typo in contact
1 file changed, 1 insertion(+)
create mode 100644 contact.html
```

```
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git status
interactive rebase in progress; onto 951a039
Last commands done (2 commands done):
   pick e2760a1 fix: typo in contact
   pick 4bb9e79 fix: typo in contact
No commands remaining.
You are currently rebasing branch 'main' on '951a039'.
  (all conflicts fixed: run "git rebase --continue")

nothing to commit, working tree clean
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git rebase --continue
Successfully rebased and updated refs/heads/main.
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 283 bytes | 141.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/natavann/Cafe-Intermediate.git
   951a039..be110d6  main -> main
```

```
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git checkout feature/style-colours
error: pathspec 'feature/style-colours' did not match any file(s) known to git
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git checkout develop
error: pathspec 'develop' did not match any file(s) known to git
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git checkout -b develop
Switched to a new branch 'develop'
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git checkout -b feature/style-colours
Switched to a new branch 'feature/style-colours'
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git stash apply stash@{0}
On branch feature/style-colours
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   styles.css

no changes added to commit (use "git add" and/or "git commit -a")
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git add styles.css && git commit -m "feat: finish colour tweak"
[feature/style-colours f0cac98] feat: finish colour tweak
 1 file changed, 1 insertion(+)
```

```
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git push -u origin feature/style-colours
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 302 bytes | 302.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature/style-colours' on GitHub by visiting:
remote:      https://github.com/natavann/Cafe-Intermediate/pull/new/feature/style-colours
remote:
To https://github.com/natavann/Cafe-Intermediate.git
 * [new branch]      feature/style-colours -> feature/style-colours
branch 'feature/style-colours' set up to track 'origin/feature/style-colours'.
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git reset --soft HEAD~1
```

Exercise 5 — Undo Mis-commit with `reset`

```
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git reset --soft HEAD~1
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git log --oneline -n 3
be110d6 (HEAD -> feature/style-colours, origin/main, main, develop) fix: typo in contact
951a039 Merge pull request #4 from natavann/text-change-B
b587165 (origin/text-change-B) Merge branch 'main' into text-change-B
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git reset --hard HEAD~1
HEAD is now at 951a039 Merge pull request #4 from natavann/text-change-B
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git push --force
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/natavann/Cafe-Intermediate.git
 + f0cac98...951a039 feature/style-colours -> feature/style-colours (forced update)
```

Exercise 6 — Version Tagging

```
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git pull
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 4 (delta 1), reused 3 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 1.15 KiB | 588.00 KiB/s, done.
From https://github.com/natavann/Cafe-Intermediate
   be110d6..acd1472  main       -> origin/main
Updating be110d6..acd1472
Fast-forward
 styles.css | 1 +
 1 file changed, 1 insertion(+)
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git tag -a v0.1 -m "First public demo"
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git push origin v0.1
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 162 bytes | 162.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/natavann/Cafe-Intermediate.git
 * [new tag]         v0.1 -> v0.1
```

```
natavan@DESKTOP-GSHQS04:~/cafe-intermediate$ git stash list
stash@{0}: On main: wip: colour tweak
stash@{1}: On main: wip: colour tweak
```

```
*   commit acd147204ec1acb3a350db3935c9d82733b2e2a4 (HEAD -> main, tag: v0.1, origin/main)
|\  Merge: be110d6 f0cac98
| | Author: natavann <87769643+natavann@users.noreply.github.com>
| | Date:   Tue Mar 31 17:45:25 2026 +0200
| |
| |     Merge pull request #5 from natavann/feature/style-colours
| |
| |     feat: finish colour tweak
| |
| * commit f0cac9877f72e41d151f7da64f3c9f4f447101c8
|/  Author: natavan <natavan.elte@gmail.com>
|   Date:   Tue Mar 31 17:40:51 2026 +0200
|
|       feat: finish colour tweak
|
* commit be110d63f7b7753ab4e6dca41473edfeadca7622 (develop)
| Author: natavan <natavan.elte@gmail.com>
| Date:   Tue Mar 31 17:12:14 2026 +0200
|
|     fix: typo in contact
|
*   commit 951a03987896f943985b71c2ba45e0f2972cfe36 (origin/feature/style-colours, feature/style-colours)
|\  Merge: 3d7f268 b587165
| | Author: natavann <87769643+natavann@users.noreply.github.com>
| | Date:   Tue Mar 31 16:55:46 2026 +0200
| |
| |     Merge pull request #4 from natavann/text-change-B
| |
| |     feat: slogan variant B
| |
| *   commit b587165242d0683bbc75c9653578017d7a809eea (origin/text-change-B)
| |\  Merge: f014ac3 3d7f268
| |/  Author: natavann <87769643+natavann@users.noreply.github.com>
|/|   Date:   Tue Mar 31 16:54:22 2026 +0200
| |
| |       Merge branch 'main' into text-change-B
| |
*   |   commit 3d7f26825df08a79cc004fb4839784c4396411de
|\ \  Merge: 2221e89 cbe16b2
| | | Author: natavann <87769643+natavann@users.noreply.github.com>
| | | Date:   Tue Mar 31 16:43:40 2026 +0200
| | |
| | |     Merge pull request #3 from natavann/text-change-A
```

We use stash for short live context switches when the code is messy or broken , not worth a commit. Temporary branch is for longer tasks or unfinished work within a team for feedback. Reset hard is dangerous because it permanently deletes any uncommitted changes and rewrites history by discarding commits.